Integral selection of a spatially varying negative-Einstein calibration

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1. Statement

Let (M,g) be a closed oriented smooth four-manifold. Let ω be a smooth self-dual two-form with $|\omega|^2 = 2$, and let t be a smooth positive function satisfying $0 < m \leqslant t \leqslant M_0 < \infty$. Use exterior norms for forms and Hilbert–Schmidt norms for Weyl endomorphisms. Set

$$A = t\left(\frac{2}{3}I - \omega \otimes \omega\right), \qquad R = W^+ - A, \qquad E = \mathrm{Ric} + g,$$

$$d = \fint (|E|^2 + |R|^2), \qquad H_2 = \fint |\nabla^2 A|^2. \tag{1}$$

Theorem 1. There is a constant depending only on m, M_0 such that

$$\boxed{\fint \left[|dt|^2 + (t - \tfrac{1}{2})^2 + |\nabla\omega|^2\right] \leqslant C_{m,M_0}\left(d + \sqrt{dH_2}\right).} \tag{2}$$

The conclusion does not assume small amplitude variation. No Sobolev constant, injectivity radius, volume lower bound, scalar spectral gap or bound on W^- appears. The second-order budget in (1) is explicit.

The statement also holds when only the unoriented real line spanned by ω is globally defined. Pass to its orientation double cover; A, t, d, H_2 and all energy densities descend, and normalized integrals are unchanged. Thus the assertion does not assume that a global Kähler orientation of this line has already been chosen.

The theorem extends constant-amplitude compatibility in a specific direction: the amplitude is a function, and its variation is part of the conclusion. It does not classify general Weyl tensors. A global nonvanishing rank-one model and the displayed spectral bounds are genuine inputs. It supplies neither global integrability nor an exact Einstein repair when $d > 0$.

2. The sourced conformal identity

Put

$$u = t^{1/3}, \quad v = \log u, \quad \widehat{g} = u^2 g, \quad \Omega = u^2\omega,$$

$$Q_X = \nabla_X\omega - dv \wedge \iota_X\omega + X^\flat \wedge \iota_{\nabla v}\omega, \qquad P = d\omega + 2dv \wedge \omega. \tag{3}$$

The conformal connection formula gives $\widehat{\nabla}_X\Omega = u^2 Q_X$ and $d\Omega = u^2 P$. In particular $|\Omega|^2_{\widehat{g}} = 2$.

Let $J = \operatorname{div} W(A)$, where $W(A)$ is the Weyl four-tensor associated with A. The first-jet identity proved in the companion variable-calibration calculation is

$$|J|^2 = 2t^2|Q|^2 + 4t^2|P|^2. \tag{4}$$

Its conformal kernel is essential: estimating $\nabla\omega$ from J before including dt would miss this kernel.

The weak contracted Bianchi identity gives, for any smooth test tensor Z,

$$\left|\int \langle J, Z\rangle\right| \leqslant C(\|E\|_2 + \|R\|_2)\|\nabla Z\|_2. \tag{5}$$

This is integration by parts; it requires no bound on ∇E. Taking $Z = J$, and using the parallel algebraic map $A \mapsto W(A)$,

$$\oint |J|^2 \leqslant C\sqrt{dH_2}, \qquad \oint |Q|^2 \leqslant C_m\sqrt{dH_2}. \tag{6}$$

All objects are smooth on the fixed closed manifold, so the test is legitimate without a limiting regularity argument.

3. A coercive scalar equation without assuming Kähler geometry

We now turn (6) into control of du. This step uses compactness of the manifold only for integration by parts, not a compactness theorem for a family of metrics.

Let $\Delta = d^*d$ be nonnegative and write $s = -4 + e$, where $|e| \leqslant 2|E|$. Set

$$\zeta = \langle R\omega, \omega\rangle, \qquad |\zeta| \leqslant 2|R|.$$

Scalar curvature and Weyl curvature transform as

$$\hat{s} = u^{-3}(6\Delta u + su), \qquad \langle \widehat{W}^+\Omega, \Omega\rangle_{\hat{g}} = -\frac{8}{3}u + u^{-2}\zeta. \tag{7}$$

No complex-analytic identity has been used here.

Test the self-dual Weitzenböck equation against $f\Omega$, for a smooth real function f. Because $|\Omega|^2_{\hat{g}} = 2$, the norm derivative term vanishes. Since Ω is self-dual, the exterior and codifferential cross terms coincide. Thus

$$\begin{aligned}\int_{\hat{g}} f(\hat{s} + 8u) = &\int_{\hat{g}} f\left(3|d\Omega|^2 - \tfrac{3}{2}|\widehat{\nabla}\Omega|^2\right)\\ &+ 3\int_{\hat{g}}\langle df \wedge \Omega, d\Omega\rangle + 3\int_{\hat{g}} f u^{-2}\zeta.\end{aligned} \tag{8}$$

Equation (8) holds for an arbitrary almost-Hermitian form of constant norm; closedness and integrability are not assumed.

Let $u_0 = 2^{-1/3}$, and choose

$$f = \frac{u - u_0}{u}. \tag{9}$$

Using $d\mu_{\hat{g}} = u^4 d\mu_g$, the left side of (8) is

$$6\int |du|^2 + 8\int u(u^3 - u_0^3)(u - u_0) + \int e\,u(u - u_0). \tag{10}$$

The potential in this expression is coercive:

$$u(u^3 - u_0^3)(u - u_0) = u(u^2 + uu_0 + u_0^2)(u - u_0)^2 \geqslant m^{1/3}u_0^2(u - u_0)^2. \tag{11}$$

Here are bounds for every term on the right of (8). The first is at most $C_{m,M_0}\int |Q|^2$ in absolute value, because

$$|\widehat{\nabla}\Omega|^2_{\hat{g}} = u^{-2}|Q|^2, \quad |d\Omega|^2_{\hat{g}} = u^{-2}|P|^2, \quad |P|^2 \leqslant 3|Q|^2.$$

For a unit normalized self-dual form, $|\xi \wedge \Omega|_{\hat{g}} = |\xi|_{\hat{g}}$. Since $df = u_0u^{-2}du$, the cross term in (8) is bounded by

$$3u_0\int |du||P| \leqslant 3\int |du|^2 + C\int |Q|^2. \tag{12}$$

Finally, combine the Weyl-error term and the last term in (10):

$$\left|\int u(u-u_0)(3\zeta - e)\right| \leqslant 4m^{1/3}u_0^2 \int (u-u_0)^2 + C_{m,M_0}\int (|R|^2 + |E|^2). \tag{13}$$

Equations (8)–(13) therefore yield

$$⨍ \left[|du|^2 + (u-u_0)^2\right] \leqslant C_{m,M_0}\left(⨍ |Q|^2 + d\right). \tag{14}$$

No Poincaré inequality is used. The restoring term is the nonlinear scalar-curvature potential, not a scalar spectral gap.

Since $t = u^3$ and u stays in a fixed positive compact interval, $|dt| \leqslant C|du|$ and $|t - 1/2| \leqslant C|u - u_0|$. Equation (3) also gives $|\nabla\omega|^2 \leqslant C(|Q|^2 + |du|^2)$. Insert (6) into (14) to prove (2).

4. The opposite sign of the variable amplitude

The sign of the amplitude is not silently discarded. Suppose instead $-M_0 \leqslant t \leqslant -m < 0$, retaining (1). Set $u = |t|^{1/3}$, so that the Weyl term in (7) is now $+8u/3 + u^{-2}\zeta$. The left side of (8) becomes $\int_{\widehat{g}} f(\hat{s} - 8u)$. Choose $f = u^{-2}$. Its expression in the original metric is

$$-6\int |d\log u|^2 - 4V - 8\int u^3 + \int e.$$

The right side is

$$3\int |P|^2 - \tfrac{3}{2}\int |Q|^2 - 6\int \langle d\log u \wedge \omega, P\rangle + 3\int \zeta.$$

Use $6ab \leqslant 3a^2 + 3b^2$, and cancel the resulting exterior energy term. This gives the explicit sign obstruction

$$3⨍ |d\log u|^2 + 4 + 8⨍ |t| \leqslant \tfrac{3}{2}⨍ |Q|^2 + \sqrt{40d}.$$

Together with (6), it proves

$$\boxed{(1 + H_2)d \geqslant c_m > 0.} \tag{14a}$$

The upper amplitude bound is only needed when replacing H_2 by the alternative budget below. Thus a uniformly nonzero negative amplitude is incompatible with negative-Einstein approximation at vanishing joint error, even when the amplitude varies spatially.

5. Input alternatives and exact limit

Set $\varphi = \sqrt{t}\,\omega$, so that

$$A = \frac{|\varphi|^2}{3}I - \varphi\otimes\varphi.$$

The bounded-section Hessian estimate and its fourth-power-gradient consequence give

$$H_2 \leqslant C_{M_0}(1 + B_\varphi + d), \qquad B_\varphi = ⨍ |\Delta_H\varphi|^2. \tag{15}$$

The self-dual connection sees $W^+, s, \mathrm{Ric}^\circ$, not W^-. Consequently an alternative statement of (2) is

$$⨍ \left[|dt|^2 + (t - \tfrac{1}{2})^2 + |\nabla\omega|^2\right] \leqslant C_{m,M_0}\left(d + \sqrt{d(1 + B_\varphi + d)}\right). \tag{16}$$

Amplitude derivatives are included in $\Delta_H(\sqrt{t}\,\omega)$; they are not assumed small or suppressed by notation.

For a sequence, the sufficient condition is $d \to 0$ and $dH_2 \to 0$, not necessarily bounded H_2. Equation (2) then forces integral stabilization of both the amplitude and the line direction. If $d = 0$ for a fixed smooth metric, (2) gives $t = 1/2$ and $\nabla\omega = 0$; the orientation cover is negative Kähler–Einstein. This exact conclusion is classical, not a new classification theorem.

When W^+ has simple lowest eigenvalue $\lambda < 0$, take its unoriented eigenline and $t = -3\lambda/4$. If the other two eigenvalues are λ_2, λ_3, then

$$|W^+ - A|^2 = \frac{1}{2}(\lambda_2 - \lambda_3)^2. \tag{17}$$

Thus (2) has a curvature-spectral interpretation on the actual line cover. A uniform gap from zero and a genuine simple eigenline are still required. The joint derivative budget is not supplied by Ricci pinching alone.

6. Relation to the research program and classical theory

The constant-parameter criterion is in `round21_analytic.md`. The new conformal first-jet identity is proved in the companion `line1_variable_calibration.md`. The present argument adds the weighted scalar test (9): on the negative branch it removes the need to prescribe constant amplitude before applying differential compatibility.

Derdziński's exact conformal-Kähler theorem is the relevant classical comparison: *Self-dual Kähler manifolds and Einstein manifolds of dimension four*, Compositio Math. 49 (1983), 405–433, original article. The zero-source conformal kernel and the exact negative-Einstein conclusion belong to that tradition. The present statement is an integral stability estimate with explicit residual and derivative budgets; exhaustive publication-priority verification is not asserted.

For Yau26 input, $\fint |\mathrm{Ric} + g|^2 \leqslant 4\delta^2$. One must additionally establish the half-Weyl spectral conditions and the joint budget. The theorem does not claim they follow for every near-Einstein metric, and does not infer an exact metric from integral parallelness alone.